Problem Set 3

First Order Linear Differential Equations

PSet Tips

- Remember to explain the reasoning behind all of your answers.
- Keep your work organized and assign a page to each problem on Gradescope.
- You don't have to do this by yourself — office hours, the MQC, and your fellow students are great resources if you get stuck.

In problems 1 and 2, find the general solution to the differential equations.

1 $(x^2+1)\dfrac{dy}{dx} + 2xy - 3x = 0$

2 $xy' + y = 4\sqrt{x}\,; \qquad x > 0.$

In problems 3–7, solve the initial value problems.

3 $\dfrac{dy}{dx} = -2xy + 2x\,e^{(-x^2)}\,; \qquad y(0) = 2.$

4 $xy' = 2y + x^3 \sin(x)\,; \qquad y(1) = 0.$

5 $(x^2 + 1)y' + 5xy = x\,; \qquad y(0) = 2.$

6 $xy' + 4y = \dfrac{e^{-x}}{x^3}\,; \qquad y(1) = 0,\ x > 0.$

7 $(1 + x)y' + y = \cos(x)\,; \qquad y(0) = 1,\ 1 + x > 0.$